A069 v0.6.1

Core separation & subgrid vortex-line tracking repair | 19 October 2026 | DEVELOPMENT - NOT FROZEN, NOT CERTIFIED

Abstract

A069 v0.6.1 repairs two distinct weaknesses in v0.5.0/v0.6.0: the use of lattice-face centers as vortex-line positions and the absence of a strict core-separation plus mesh-resolution admission gate. We introduce bilinear complex-field zeros for each wound plaquette, oriented dual-graph polylines and subcell Fourier-grid offsets. The previous GP evolution code remains byte-pinned and unmodified. These are numerical diagnosis improvements, not validation of a stable particle knot or Swirl Clock.

Scientific and model scope

The simulated medium solves the dimensionless, compressible Gross-Pitaevskii equation. It is neither a resolved incompressible Euler trefoil nor a physical atomic clock. The historical Fourier A and Torus B curves are geometry inputs. No golden-ratio, Compton, Lorentz, particle-mass or desired clock-frequency calibration is used.

Method and fail-closed readiness

The scanner evaluates nearby, nonadjacent line-segment pairs while excluding 7% of the periodic parameterized arclength neighborhood. Its distance is a numerical nonlocal screening diagnostic - not mathematically certified reach or physical core thickness. We reject readiness if the GP core has fewer than four grid cells per xi, if xi / screened nonlocal gap exceeds 0.30, or if xi times maximum discrete centerline curvature exceeds 0.30. Thresholds are preregistration proposals for a future production campaign, not universal laws.

A face carrying integer circulation is localized by solving both Re(Psi)=0 and Im(Psi)=0 in the bilinear interpolated plaquette. Zeros on shared face edges are marked boundary-degenerate rather than falsely connected into unique lines. The face crossing points are connected through a dual graph only when each node has one incoming and one outgoing link. This is a geometry diagnostic and not a knot invariant or ambient-isotopy certificate.

Local numerical results (executed)

GP case	Gap d	xi/d	xi kmax	xi/dx	Pre / post graph	G32
A_N32	0.284	1.478	0.703	1.34	1 / 1	FAIL
B_N32	1.137	0.370	0.430	1.34	1 / 1	FAIL
B_N48	1.137	0.370	0.430	2.02	2 / 1	FAIL

All three quick preflights use xi=0.42, box=10, dt=0.006 and only eight time steps. They deliberately fail the core/grid qualification conditions. Initial/final component counts are not reconnection events.

Decisive aliasing negative control

For B on 48^3, the SAME INITIAL GP FIELD, before physical time evolution, yields different dual-graph component counts under two subcell Fourier translations: offset [0.37, 0.29, 0.41] -> 2 components, offset [-0.23, 0.19, -0.31] -> 1 components. As there is zero elapsed GP time, the disagreement is incontrovertibly an extraction/grid-alignment effect rather than a dynamical reconnection. This specifically invalidates topology promotion from component count alone.

Results and epistemic gates

Gate	Executed result	Permitted statement
G32 core/readiness	FAIL (all 3 preflights)	No dynamics qualification

Gate	Executed result	Permitted statement
G33 subgrid zeros	Mixed geometric diagnostics	Software localization implemented
G34 knot class and topology transport	UNRESOLVED	No reconnection determination
G35 independent intrinsic eigenmode	UNRESOLVED	No Swirl Clock frequency

There are 23 passing local regression tests (9 inherited v0.6.0 and 14 new). Neither an official v1.0.9 FREEZE/BASIC nor any native Arc A770 backends or source-independent physics holdout has been executed for v0.6.1.

Next production qualification

A useful candidate is B with xi=0.28 and box=10, requiring at least approximately N=143 for >=4 cells/xi, before additional domain, dx, dt, core and phase-alignment sweeps. This candidate is NOT already validated. Fourier A at the measured 0.284 nonlocal gap would require xi no greater than approximately 0.085 under the same ratio rule, implying N roughly 470 at box=10. Memory pressure and physical core overlap must therefore be handled explicitly. If a knot reconnects in fully converged GP, record that negative result; never repair it by post-hoc parameter selection.

Provenance and limitations

Pinned upstream public v0.5.0 ZIP SHA-256: bb8e1d392e3e72c0c2caee22df406469f9f3de5504296da45cedc6fda8d04b6e. All outputs remain DEVELOPMENT; source hashes and executable diagnostics are included in the accompanying ZIP. Old KnotPlot-relaxed geometry is excluded. The two historical trefoils and mirrors are not independent physical providers.